Early Interventions for Children With Developmental Disabilities: Global Causal Evidence and Implications for Türkiye

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BA840 Using Evidence to Improve Teaching and Learning

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Review Question, Definitions, and Context

Türkiye's Ministry of National Education (MoNE) needs causal evidence to identify which early interventions (ages 0–5) most effectively improve outcomes for children with developmental delays/disabilities (DD/D). Despite legislative provisions since the 1980s, Türkiye lacks a coordinated early intervention system, standardized assessments, and sufficient trained personnel (Diken et al., 2012). Evidence-based prioritization is essential for resource allocation and national scaling decisions.

DD/D includes individuals with either a formal disability diagnosis or documented eligibility for disability services. Findings are noted where they differ between high-income and middle-/low-income contexts, though evidence is limited. Outcomes are grouped into two levels:

1. **Level 1 (preferred)**: Later educational outcomes, including K–5 reading, literacy, math, numeracy, overall achievement or test scores, and the intensity of special education services.
2. **Level 2 (secondary)**: Early outcomes used only when Level 1 data are not available, including pre-kindergarten measures of language, cognitive skills, and social-emotional/personal development.

Methods

The search included ERIC, PubMed, Elicit, Google Scholar, HOLLIS, 3ie, the World Bank Open Knowledge Repository, WWC, and EEF. Additional searches of MoNE/OECD/UNESCO/UNICEF/IEA sources were conducted for context and system descriptions only (excluded from the causal synthesis). Grey literature was retained for contextual background but excluded from the causal synthesis. Coverage extended from 2000–present, in English and Turkish, globally. Country income classification (HIC/UMIC/LMIC) was recorded to support cross-context comparisons where evidence permitted. The population comprised children prior to kindergarten entry (≤71 months at baseline) with diagnosed DD/D or eligible for EI/ECSE.

a) **Search Strings**: English and Turkish, ("early intervention" OR "early childhood special education") AND ("developmental delay" OR disability OR "special needs") AND (literacy OR numeracy OR achievement OR "school readiness"). ("erken müdahale" OR "erken çocukluk özel eğitimi") AND ("gelişimsel gecikme" OR "özel gereksinim") A shortened English string was used when databases limited query length or initial searches returned no results. Full strings, result counts, and dates are in Appendix A.
b) **Eligibility criteria:** Included studies involved pre-kindergarten children with diagnosed or EI/ECSE-eligible DD/D; EI or ECSE interventions including parent-mediated models, education-relevant outcomes, and RCT or quasi-experimental designs with comparison groups (English/Turkish, 2000–present). Excluded post-kindergarten samples, at-risk-only populations, studies without education outcomes or comparison groups, and opinion pieces.
c) **Screening:** Retrieved records were exported to Excel spreadsheets, and duplicate entries were identified and removed manually. Two-stage screening applied inclusion criteria: first at title/abstract level, then at full-text level for potentially relevant studies. Single-reviewer screening was conducted, which increased the risk of false inclusion or exclusion of studies. Due to time constraints, 200 of 887 PubMed records were screened by title; the remaining records were not reviewed, which may have resulted in missed

eligible studies. No pre-registration of the review protocol was conducted, as per course assignment parameters; however, eligibility criteria were clearly established and documented before the screening process began.

Evidence Map

Table 1. Primary causal studies (k=3)

Study	Country (Income Level)	Sample	Design	Outcome (Level)	Effect Size	Subgroup Findings	Key Implementation Features	Causal Interpretation
Stingone et al. (2026)	USA (HIC)	NYC birth cohort; N=214,370; EI recipients n=13,022; ages birth–3	Non-randomized; statistical matching on demographic and health characteristics	Level 1: Grade 3 ELA and math scores	ELA: +0.045 SD (95% CI [0.021, 0.069]); grade-level standards: ELA ratio 1.09; math ratio 1.08	Larger effects for children later needing special education (ELA ratio 1.28 vs. 1.09)	Children received EI services as part of routine practice; content, dosage, and provider were not controlled	Non-randomized; children who received EI may have differed from those who did not in ways the statistical matching could not fully account for

Sullivan & Field (2013)	USA (HIC)	National cohort (ECLS-B); preschool special education ages 3–5; kindergarten assessment ~2006–07	Non-randomized; statistical weighting to balance treatment and comparison groups	Level 1: Kindergarten reading and math scores	Reading: −0.21 SD; Math: −0.29 SD	Results were not broken down by disability type or service intensity; the overall negative finding likely looks different across different groups of children	Children received services as part of routine practice; what was delivered, how often, and by whom was not recorded	Children who received services had more severe needs to begin with — the negative scores most likely reflect that difference, not that services made things worse
Karaaslan et al. (2013)	Turkey (UMIC)	19 mother–child dyads; diagnosed disability; pre-kindergarten age	RCT: Responsive Teaching + standard services vs. standard services only; 6-month follow-up	Level 2: Developmental Quotient (DQ); maternal responsiveness; child engagement	DQ improved 42% in RT group vs. 7% in control group over 6 months	Consistent improvement across language and personal-social domains	Biweekly sessions; blinded assessment; fidelity monitored	Strongest causal warrant: small N means wide uncertainty; no later academic outcomes

Synthesis

Three studies met eligibility criteria: a Turkish RCT assessing short-term developmental outcomes (Karaaslan et al., 2013), a U.S. administrative cohort assessing Grade 3 outcomes (Stingone et al., 2026), and a U.S. national cohort assessing kindergarten outcomes (Sullivan &

Field, 2013). The studies fall into two intervention categories: parent-mediated developmental coaching (Karaaslan et al., 2013) and system-level EI/ECSE service receipt (Stingone et al., 2026; Sullivan & Field, 2013). The evidence base is small, largely U.S.-based, and heterogeneous; studies assessed different concepts at different time points, preventing quantitative synthesis. The planned HIC vs. UMIC/LMIC comparison could not be conducted due to insufficient evidence, a gap in the evidence base, not a finding.

Patterns

Karaaslan et al. (2013) evaluated Responsive Teaching (RT) added to standard services over six months. Children receiving RT showed substantially larger developmental gains than controls. Because families were randomly assigned, RT is the most likely explanation for these differences. Language and personal–social domains also improved. The study groups were very small, so the exact size of improvement is uncertain. No later academic outcomes were assessed.

Stingone et al. (2026) found EI recipients performed slightly better in Grade 3 ELA and were more likely to meet grade-level standards. For MoNE, a key result is stronger effects among children later identified for special education (ELA ratio 1.28 vs. 1.09 overall), suggesting higher returns for children with greater developmental need.

Sullivan & Field (2013) reported lower kindergarten reading (−0.21 SD) and math (−0.29 SD) scores among service recipients. This most likely reflects the fact that children who received services started out with significantly greater needs than those who did not — and no statistical method can fully correct for that difference. Services may also target developmental goals that standard academic tests simply do not measure. These findings should not be interpreted as evidence that services are harmful.

Karaaslan et al. (2013) has the strongest internal validity through random assignment, though the small sample limits precision. Both U.S. studies are vulnerable to unmeasured differences between groups that statistical methods cannot fully correct. Generalizability is limited across all three studies: findings come from two rehabilitation centers, a single city, and a service system structurally different from Türkiye's. Fidelity was monitored only in Karaaslan et al.

Distinguishing Correlation from Causation

Only Karaaslan et al. (2013) randomly assigned participants, so its findings support the strongest cause-and-effect conclusions. The two U.S. studies used statistical methods to make the groups more comparable, but these methods cannot fully account for differences that were never measured, such as how severe a child's needs were or how actively a family sought out services. Those non-randomized results should be read as associations that point in a direction, not as definitive proof of cause and effect.

Limitations

With only three studies meeting eligibility criteria, findings cannot be generalized, and no formal risk-of-bias tool was applied. Reference lists of included studies were not hand-searched, representing a potential source of missed studies. No Turkish-language causal studies were identified beyond Karaaslan et al. (2013), which likely reflects gaps in database indexing of Turkish-language literature rather than an absence of relevant research.

Implications for MoNE

Defensible now: Parent-mediated RT coaching can improve short-term developmental outcomes for children with DD/D in Türkiye under trial conditions (Karaaslan et al., 2013). US evidence suggests that receiving EI before age 3 is associated with small positive Grade 3 outcomes, with stronger signals for higher-need children (Stingone et al., 2026). However, differences between the US IDEA/Part C system and Türkiye's rehabilitation-center model limit direct policy transfer.

Not defensible yet: Naming a single best model to sustain K–5 gains in Türkiye; assuming early gains will persist into later academic outcomes; or applying US effect sizes to the Turkish context.

Actionable next steps

Türkiye should prioritize expanding RT and similar parent-mediated programs with outcome evaluation built into implementation from the outset. Linking early service records to later school data would allow MoNE to determine whether early gains translate into sustained academic performance. Evidence from the broader literature on fade-out in comparable country contexts indicates that early intervention effects tend to diminish over time when support is discontinued (Jeong et al., 2021). This suggests that program enrollment alone is insufficient; continued support through the transition into formal schooling is essential to preserving early gains.

Appendix A. Search Log

Part 1: Systematic Search Runs

Date	Database	Run	Exact Query / Strategy	Filters Applied	Records Retrieved	Notes

3/2/2026	ERIC	1	("early intervention" OR "early childhood special education") AND ("developmental delay" OR disability OR "special needs") AND (literacy OR numeracy OR achievement OR "school readiness")	2000–present; EN/TR	40	9 relevant to context; 6 in age group; 2 with needed data
3/5/2026	ERIC	2	("early intervention" OR "early childhood special education") AND ("developmental delay" OR "special need")	2000–present; TR	0	Turkish core string; no findings
3/5/2026	ERIC	3	("early intervention" OR "early childhood special education") AND ("developmental delay" OR disability OR "special needs") AND (literacy OR numeracy OR achievement OR "school readiness")	Filtered by age group and disabilities; 2000–present; EN/TR	12	Literature review only; no causal studies

3/10/2026	OECD / UNICEF / IEA	1	Browsed portals for context and system-description information	2000–present; EN/TR	Reports reviewed	Context/system-description only; excluded from causal synthesis
3/15/2026	OECD / UNICEF / IEA	2	Browsed portals for context and system-description information	2000–present; EN/TR	Reports reviewed	Context/system-description only; excluded from causal synthesis
3/16/2026	OECD / UNICEF / IEA	3	Browsed portals for context and system-description information	2000–present; EN/TR	Reports reviewed	Context/system-description only; excluded from causal synthesis
3/19/2026	Elicit	1	Early intervention services developmentally disabled children Turkey	2000–present; EN/TR	30	Systematic run, 5 relevant sources used.
3/19/2026	3ie Development Evidence Portal	1	Browsed portal	Filtered by: Education sector; Equity Dimension / Disability	2	No relevant findings.
3/20/2026	PubMed	1	("early intervention" OR "early childhood special education") AND ("developmental delay" OR "special need")	2000–present; TR	12	Turkish core string; 1 relevant finding
3/20/2026	PubMed	2	("early intervention" OR "early childhood special	2000–present; EN/TR	887	Stopping rule applied: 200 titles screened before threshold

			education") AND ("developmental delay" OR disability OR "special needs") AND (literacy OR numeracy OR achievement OR "school readiness")			reached. 6 relevant sources used.
3/20/2026	World Bank Open Knowledge Repository & Data Portal	1	Browsed datasets and reports; indicators: education, social development; EI services, disabilities, early intervention	Country, region, income level (high/middle/low); 2000–present	Datasets and reports reviewed	Context and measurement only; no causal studies identified
3/21/2026	Elicit	2	Early intervention services and academic outcomes for developmentally disabled young children	2000–present; EN/TR	10	Systematic run, 4 relevant sources used
3/23/2026	OECD / UNICEF / IEA	4	Browsed portals for context and system-description information	2000–present; EN/TR	Reports reviewed	Context/system-description only; excluded from causal synthesis
3/24/2026	WWC	1	Browsed portal	Filtered by: Children and Youth with Disabilities → filtered by Preschool	176 → narrowed to 36	4 sources used

3/24/2026	UNESCO	1	("early intervention" OR "early childhood special education") AND ("developmental delay" OR disability) + UNESCO	2000–present; EN/TR	Relevant reports reviewed	Context and measurement only; no causal studies identified
3/26/2026	Google Scholar	1	("early intervention" OR "early childhood special education") AND ("developmental delay" OR disability OR "special needs") AND (literacy OR numeracy OR achievement OR "school readiness")	2000–present; EN/TR	10	Stopping rule: after 10 consecutive non-relevant abstracts or 40 consecutive non-relevant titles
3/26/2026	3ie Development Evidence Portal	2	Browsed portal	Filtered by: Education sector; Country; UN Global Goals / Quality of Education; Equity Dimension — Education and Disability	5	4 non-relevant; 1 used
3/27/2026	World Bank Open Knowledge Repository & Data Portal	2	Browsed datasets and reports; indicators: education, social development	Country, region, income level; 2000–present	Datasets and reports reviewed	Context and measurement only

3/27/2026	HOLLIS	1	("early intervention" OR "early childhood special education") AND ("developmental delay" OR disability OR "special needs") AND (literacy OR numeracy OR achievement OR "school readiness")	Filtered by Turkish-language publications, age group; 2000–present	107	4 relevant sources identified
3/27/2026	EEF Evidence and Resources Portal	1	Browsed portal	Filtered by: Early Years	2	2 sources used for literature review
3/27/2026	EEF Evidence and Resources Portal	2	Browsed portal	Filtered by: Special Educational Needs	4	2 used for literature review
3/27/2026	EEF Evidence and Resources Portal	3	Browsed portal	Filtered by: Early Literacy	2	Does not meet DD/D eligibility criteria
3/28/2026	Elicit	3	Early intervention services developmentally disabled children Turkey OR Early intervention services and academic outcomes for developmentall	2000–present; EN/TR	10	Systematic run. 3 relevant sources identified and used.

			y disabled young children			
3/28/2026	OECD / UNICEF / IEA	5	Browsed portals for context and system-description information	2000–present; EN/TR	Reports reviewed	Context/system-description only; excluded from causal synthesis
Multiple (≥20 sessions)	MoNE (meb.gov.tr)	—	Strictly Turkish-language browsing of Ministry of Education website	Turkish only; 2000–present	≥20 browsing sessions	Context/system-description only; no causal data available
3/30/2026	Elicit AI	8	Disability access inequities in Türkiye; EI services ages 0–5; global causal evidence; LMIC evidence; long-term outcomes (5 queries per round; 4 queries Round 6)	N/A	390 total (before de-duplication)	Context/system-description only. Background information, literature, data.

Appendix B. Exclusion Log

ID	Study	Stage	Primary Exclusion Reason	Notes
1	Karaaslan, Diken, & Mahoney (2011) RT pilot	Full text	Not causal	Two-case pre/mid/post design; no comparison group
2	Stingone et al. (2022) EI and Grade 3 tests among lead-exposed	Full text	Wrong population	Lead-exposed at-risk cohort; not DD/D-diagnosed or eligible

3	Koç Yekedüz et al. (2025) family-centeredness	Full text	Not causal	Cross-sectional service quality and satisfaction survey; no comparison group
4	Bayhan & Sipal (2011) SE Türkiye family perspectives	Full text	Not causal	Service delivery perceptions; no impact estimates
5	Diken et al. (2012) ECI/ECSE system overview Türkiye	Full text	Not causal	System overview; no effect estimates
6	Rosenberg et al. (2008) prevalence/participation	Full text	Not causal	Participation and coverage analysis; no impact estimates
7	Hastings et al. (2012) LMIC PDD interventions review	Full text	Wrong publication type	Systematic review; used for reference mining only
8	Halfon et al. (2009) Commonwealth Fund report	Full text	Not causal	Systems comparison report; no impact estimates
9	Froiland (2021) Early intervention programs. EBSCO Research Starters.	Title/abstract	Narrative overview — no empirical data identifiable from abstract	No comparison group; no empirical study design
10	Chamberlin (1987)	Title/abstract	Outside date range (pre-2000)	Published 1987; outside 2000–present eligibility window
11	Campbell & Ramey (1994) Abecedarian follow-up	Title/abstract	Outside date range (pre-2000)	Published 1994; outside eligibility window
12	Leitner et al. (2007) long-term survey	Full text	Not causal	Treated-only cohort; no comparison group
13	Bailey et al. (2005) 36-month family outcomes	Full text	Not causal	Descriptive parent interview study; no comparator; family outcomes only
14	Republic of Türkiye MoFSS (2023) Autism Action Plan	Full text	Not causal	Policy/action plan document; context only

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Policy Alternatives and Evaluation Criteria for Early Intervention (Ages 0–5) for Children With Developmental Delays/Disabilities (DD/D) in Türkiye

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BA840 Using Evidence to Improve Teaching and Learning

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Türkiye Ministry of National Education (MoNE) seeks evidence-based policy to prioritize early interventions (ages 0–5) for children with developmental delays/disabilities (DD/D) to improve outcomes and reduce access inequities (Diken et al., 2012). Turkish RCT evidence supports parent-mediated Responsive Teaching (RT) (Karaaslan et al., 2013). International research highlights the need for clear service standards, data systems, and teacher support for equity and inclusion (Alkhateeb et al., 2016; Sapiets et al., 2023).

Policy Alternatives

Parent-Mediated Responsive Teaching (RT) Coaching at Scale

This model trains parents of children aged 18–60 months with DD/D, prioritizing underserved families, to use RT strategies at home, improving child communication and adaptive skills (Karaaslan et al., 2013).

Target Group: Children 18–60 months with DD/D, especially underserved families.

Delivery Model: RT coaches provide sessions through Special Education and Rehabilitation Centers; hybrid in-person and tele-coaching; supervision and fidelity checks included.

Dosage: Pilot both the Turkish RCT model (2×/week, 90 min, 4 months) and an international weekly model (weekly, 60–90 min, 6 months); select optimal dosage based on pilot data (Mahoney & MacDonald, 2007; Jeong et al., 2021).

Timeline: Months 0–6: Adapt materials, train coaches, plan measurement. Year 1: Pilot in 4–6 provinces. Years 2–3: Scale up contingent on fidelity and outcome benchmarks.

Cost Category: Medium (personnel, training, supervision, tele-support, materials).

Alternative 2: Minimum Service Package (MSP) + Registry + Family Navigation

This approach standardizes and monitors EI/ECSE services to reduce inequity and strengthen system learning (Sapiets et al., 2023; Schalock et al., 2011).

Target Group: Children 0–5 with DD/D receiving EI/ECSE, emphasis on underserved regions.

Delivery Model: Centers enforce an MSP for service, dosage, qualifications, and documentation; a digital registry tracks access/outcomes; keyworkers guide families through access and referrals.

Dosage: Tailored per child; measured and reviewed every 8–12 weeks.

Timeline: Months 0–9: Define standards, develop registry, train staff. Year 1: Pilot rollout. Years 2–4: Expand and improve system.

Cost Category: Medium (training, IT, navigation, quality assurance).

<u>Alternative 3: Transition Package with Defined Inclusion Model (Targeted Pilot)</u>

This pilot supports children with DD/D, ages 3–5, as they transition to kindergarten, ensuring inclusion is meaningful (Alkhateeb et al., 2016; Reimers, 2020).

Target Group: Children 3–5 with DD/D transitioning into kindergarten, in regions with system readiness.

Delivery Model: Itinerant ECSE coaches support preschools with teacher coaching, targeted small-group supports, ongoing professional development, and structured teacher feedback and family engagement.

Dosage: Teacher-facing: Monthly coaching cycles. Child-facing: Inclusive preschool with targeted small-group support as needed.

Timeline: Year 1: Pilot in 1–2 regions; develop workforce and tools. Years 2–5: Gradual scale-up as workforce and routines stabilize.

Cost Category: High (specialist workforce, sustained coaching, professional development).

<u>Evaluation Criteria and Measurement</u>

Criterion	**Definition**	**Measurement Approach**
Effectiveness	Improvement in child developmental and educational outcomes directly attributable to the policy	Effect size (≥0.20 SD; Coalition for Evidence-Based Policy, 2003), standardized tests, school linkage
Equity Impact	Reduced gaps by SES, region, severity	Subgroup analysis: enrollment, dosage, effect sizes
Scalability & Fidelity	Ability to implement at scale while maintaining intervention fidelity	Fidelity scores, supervision, workforce coverage, registry/audit rates, attrition
Fiscal Cost/Cost-effectiveness	Cost category and value for money relative to outcomes	Personnel, training, IT, outcome per cost category, per child cost
Political Feasibility & Relevance	Fit to Türkiye's system, stakeholder support, risk/timeline	Stakeholder feedback, complaints, risk register, time to results, sustainability
Quality of Life (QoL)	Improvement in participation and subjective wellbeing of children and families	At least one participation and one parent-reported wellbeing metric per option (Zekovic & Renwick, 2003)

Theory of Change

RT Coaching

Inputs → RT curriculum, trained coaches, supervision, tele-coaching
Activities → Parent coaching , practice routines, group sessions, fidelity checks
Outputs → Attendance, completion, fidelity, parent engagement
Short-term Outcomes → Improved parent responsiveness, interactions, child communication/adaptive skills
Long-term Outcomes → Better school readiness, learning, less special education need
Mechanisms → Responsive interactions, skill generalization, parent self-efficacy
Assumptions → Sustained parent participation, coach fidelity,access

MSP + Registry + Navigation

Inputs → MSP standards, digital registry, navigators, QA/audit
Activities → Documentation, reviews, audits, navigation, provider support
Outputs → Complete records, dosage/goals, navigation contacts, barrier data
Short-term Outcomes → Reduced service variation, better access/continuity
Long-term Outcomes → Better quality,equity, child outcomes
Mechanisms → System transparency, accountability, reduced barriers
Assumptions → Feasible documentation, enforceable standards, effective navigation

Transition Package/Inclusion Model

Inputs → ECSE coaches, transition tools, leader support
Activities → Teacher coaching, inclusive routines, PLCs, transition/family planning
Outputs → Schools/teachers supported, visits, participation, transition plans, inclusion indicators
Short-term Outcomes → Better inclusive instruction, smoother transitions
Long-term Outcomes → Improved early-grade learning, less fade-out, sustained inclusion
Mechanisms → Teacher support, continuity, embedded learning
Assumptions → School readiness, teacher time, consistent inclusion

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Policy Matrix, Recommendation, and Supporting Analysis

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BA840 Using Evidence to Improve Teaching and Learning

Dr. Emiliana Vegas, Dr. Marty West April 14, 2026

Policy Matrix

Türkiye's eligible population of children ages 0–5 with DD/D is estimated at 153,000–291,000, based on WHO GBD 2019 European Region prevalence modeling (3.4%) and Demirci & Kartal's (2016) ASQ screening study in Izmir (6.47%), applied to Türkiye's 0–5 population of approximately 4.5 million. The official disability registry (36,226 children ages 0–4) is excluded as a baseline, it counts only families who have already accessed formal hospital services (EYHGM, 2020). The Year 3 target of 8,000–10,000 children represent 3–7% coverage, a modest but honest starting point for a system that has substantial rehabilitation infrastructure but no coordinated EI program specifically for children ages 0–5 with DD/D.

Criterion	Alt 1: RT Coaching	Alt 2: MSP + Registry	Alt 3: Inclusion Model	Hybrid (Recommended)
Effectiveness Improvement in child developmental outcomes. ***Measured by effect size (≥0.20 SD), standardized tests, school linkage.***	+0.15–0.22 SD in communication and adaptive behavior by Year 3 (Karaaslan et al., 2013; Jeong et al., 2021; WHO-UNICEF, 2023). School readiness cannot be projected, no Turkish baseline.	No direct child intervention. Indirect gains through improved service consistency; effect size not quantifiable.	No Turkish effect size available (Aksu, 2022). Alt 3 generates evidence; it does not replicate it.	+0.15–0.22 SD nationally from Alt 1. Alt 2 measures outcomes but do not produce them. School readiness baseline: Year 1.
Equity Impact Reduced gaps by SES, region, severity. ***Measured by enrollment, dosage, effect size subgroup analysis.***	Free delivery through 2,505 centers in all 81 provinces. SES gap baseline unknown; Alt 2 will measure it.	First equity measurement by SES and region. Only 0.64% of children 0–6 is currently referred for evaluation (Metin et al., 2025).	Urban families only. Rural and low-SES children excluded Years 1–3 (Alkhateeb et al., 2016).	Alt 1 reaches broadly; Alt 2 measures gaps; Alt 3 bounded to 2 regions. Equity baseline: Year 1.
Scalability & Fidelity Scale while maintaining fidelity. ***Measured by fidelity scores, supervision, workforce coverage, and attrition.***	All 81 provinces via existing rehabilitation center network. Türkiye has adapted two international EI programs nationally	60+ provinces using existing staff. Turkish studies report fidelity at 60% vs. 90% in the	2 urban regions only. No national ECSE workforce exists (WHO-UNICEF,	81 provinces Alt 1; 60+ provinces Alt 2; 2 regions Alt 3. Monthly supervision; fidelity

	before (Aksu, 2022). Fidelity target: 45–55% Year 1; 65–70% Year 3.	US (Aksu, 2022).	2023); only 5 Turkish inclusion studies covering 21 children (Aksu, 2022).	dashboards from Year 2.
Fiscal Cost/ Cost-Effectiveness Value for money relative to outcomes. ***Measured by personnel, training, IT, per-child cost.***	Medium. Uses existing centers, staff, and MoF reimbursement. No new infrastructure required.	Lowest. Manual tracking system in Year 1 uses existing staff. No IT procurement Years 1–2.	Highest. Requires new specialist training pipeline.	Best overall value. Highest-cost component is bounded to 2 regions.
Political Feasibility & Relevance Fit to Türkiye's system, stakeholder support, risk, timeline. ***Measured by stakeholder feedback, risk register, time to results, sustainability.***	Very High. Results visible by Month 4–6 (Karaaslan et al., 2013). Implements an existing legal mandate, no new policy required (Aksu, 2022).	Moderate alone; essential alongside Alt 1. Addresses absence of EBP criteria in Turkish education law (Aksu, 2022).	Moderate. High cost, thin evidence base, limited reach, and institutional resistance risk.	Very High. Alt 1 delivers early wins; Alt 2 builds accountability; Alt 3 signals innovation.
Quality of Life (QoL) Improvement in participation and wellbeing. ***Measured by: ≥1 child participation and ≥1 parent-reported wellbeing measure (Renwick et al., 2003).***	Earlier identification reduces family stress (Metin et al., 2025). Parent self-efficacy gains documented (WHO-UNICEF, 2023).	Reduces navigation burden; improves access continuity (Sapiets et al., 2023).	Inclusive peer participation; reduced social isolation during kindergarten transition. *(2 regions only)*	Early family connection (Alt 1); reduced navigation burden (Alt 2); peer inclusion (Alt 3).

Recommended Hybrid Model

Türkiye should adopt three simultaneous components: parent-mediated Responsive Teaching coaching as the primary intervention (Alt 1); a Minimum Service Package, registry, and family navigation as accountability infrastructure (Alt 2); and a bounded inclusion pilot in two high-capacity urban regions (Alt 3). Direct intervention without measurement produces unverifiable outcomes; measurement without intervention produces no child gains. Both tracks must run together. This structure reflects the twin-track approach endorsed by WHO-UNICEF (2023). Two Turkish sources ground the case: Metin et al. (2025) quantifies the identification gap the hybrid addresses; Aksu (2022) establishes the legal foundation and evidence limits that define what can be scaled and what cannot.

Expected Outcomes

Learning gains. Alt 1 is projected to produce **+0.15–0.22 SD** improvement in child communication and adaptive behavior by Year 3 (Karaaslan et al., 2013; Jeong et al., 2021; WHO-UNICEF, 2023). This projection holds when coaches deliver the program as designed and parents attend at least 75% of sessions. These are gains in communication and adaptive behavior, not school readiness. One contrary finding: Sullivan & Field (2013) reported lower kindergarten scores among EI recipients, most likely because children who received services had greater needs to begin with, not because services caused harm. This reinforces the need for careful baseline measurement in Year 1.

School readiness. No projection is possible. Children with DD/D are currently identified at 49–60 months, not the 18–36 month window where Alt 1 operates (Metin et al., 2025). No cohort has been tracked early enough to generate this data. Year 1 registry establishes the baseline; Year 3 is the first defensible evidence point.

Coverage. The Year 3 target of 8,000–10,000 children represent 3–7% of the estimated eligible population of 153,000–291,000. In a system where only 0.64% of children attending pediatric clinics are currently referred for developmental evaluation (Metin et al., 2025), this is a meaningful increase.

Equity. Current EI enrollment by SES, region, or disability type is untracked. Alt 2 produces the first national measurement. Alt 1's free delivery through rehabilitation centers in all 81 provinces reaches across income levels. Gap closure targets are deferred until Year 1 data establish the baseline.

Tradeoffs

Early results vs. accountability. Alt 1 shows results fast, visible child outcomes by Month 4–6. But speed without tracking creates a blind spot: without Alt 2, there is no way to know which families are being reached and which are not. This matters because the current system already misses most children who need services, 80.8% of referrals go to child psychiatry, reflecting the most visible cases, not the full DD/D population (Metin et al., 2025). Both components must be launched together. A Month 6 check will confirm whether the right children are being enrolled.

Urban inclusion pilot vs. rural inclusion. Limiting Alt 3 to two urban regions means children in 79 provinces receive no structured inclusion support in Years 1–3, a genuine cost. Two constraints make this necessary: Türkiye has no trained itinerant coaching workforce (EYHGM, 2020), and only five Turkish inclusion studies covering 21 children exist to guide implementation (Aksu, 2022). Rural children receive Alt 1 from Year 1. Rural expansion of Alt 3 will only proceed if three conditions are met by Year 3.

Basic tracking system vs. real-time digital platform. Phase 1 uses a manual tracking system, low-cost, immediately operational, reviewed quarterly. This is accepted given budget realities. Only 60% of Turkish studies currently report implementation fidelity at all (Aksu, 2022). A functional basic system is a meaningful improvement over what currently exists.

Justification

Evidence strength. Alt 1 is grounded in the only Turkish RCT for this population (Karaaslan et al., 2013). Intellectual disability is Türkiye's largest registered disability category (20.03%; EYHGM, 2020), and developmental delay is the most common reason children ages 0–6 is clinically referred (Metin et al., 2025). Responsive Teaching was designed for children with intellectual disability and communication delays (Mahoney & MacDonald, 2007). Turkish practitioners already use similar naturalistic, play-based approaches and report consistent positive results (Aksu, 2022). US evidence adds further support: EI recipients later identified for special education showed stronger Grade 3 gains than the overall group (Stingone et al., 2026), reinforcing the case for targeting confirmed DD/D specifically.

Contextual fit. Türkiye's 2,505 rehabilitation centers serve 415,785 students nationally and have grown 90% over twelve years, the delivery infrastructure WHO-UNICEF identifies as essential for scalable EI (WHO-UNICEF, 2023). Türkiye has adapted two international EI programs nationally before: Small Steps and Portage (Aksu, 2022). Alt 1 repeats a demonstrated pattern with stronger evidence.

Feasibility under constraint. Türkiye's 2018 Special Education Services Regulation already mandates ECI for ages 0–36 months and integrated preschool for ages 36+ months (Aksu, 2022). Hybrid implements existing laws, not a new obligation. With referral rates at 0.64% and delays identified at 49–60 months, children are reaching school without skills that intervention at 18–36 months can build (Metin et al., 2025). Alt 1 alone is the strongest single alternative, but without Alt 2 its gains cannot be measured or equitably distributed. Hybrid is an integrated strategy, each component necessary for the others to work.

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Evaluation Strategy - Early Interventions for Children With DD/D in Türkiye

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BA840 Using Evidence to Improve Teaching and Learning

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Evaluation Design

Three constraints prevent a randomized controlled trial in Years 1–3. First, Türkiye's 2018 Special Education Laws guarantees EI for all eligible children ages 0–36 months; withholding services would violate legal and ethical obligations. Second, only the most visible DD/D cases are currently identified, preventing representative randomization (Metin et al., 2025). Third, the Alt 2 registry is built during Year 1, leaving no baseline data for pre-assignment randomization. This evaluation therefore uses a phased quasi-experimental design with three strategies.

Staggered provincial rollout. Provinces receiving RT coaching in Year 1 are compared to those scheduled for Year 2. Their shared national policy environment and administrative structure make this a credible comparison without randomization. Child developmental scores and, by Year 3, school readiness outcomes are compared between early-rollout and later-rollout provinces, controlling for observable province-level characteristics recorded in the Alt 2 registry.

Before-and-after comparison within provinces. This strategy assesses whether the Alt 2 registry improves system identification and service delivery. Registry activation serves as the before/after reference point within each province, with key indicators: referral rates, ÇÖZGER-to-RAM lag times, time to first service contact, and dosage compliance tracked before and after. Since the registry is the only systematic change introduced, observed improvements can reasonably be associated with its activation. One limitation is that concurrent changes, such as new policies or staffing shifts, cannot be fully ruled out. Strategy 1's cross-provincial comparison provides a partial check: consistently better indicators in registry provinces strengthen the association, though not conclusively.

Matched comparison for Alt 3. Two pilot regions are matched to two structurally comparable non-pilot regions on kindergarten enrollment rate, urban/rural classification, and disability registry density. Alt 3 is framed as a learning investment rather than an effectiveness claim. Aksu's (2022) review found Türkiye's early childhood inclusion evidence base too sparse to support national scaling without a prior pilot.

Why is a score-based eligibility comparison not yet feasible. The two stages of Türkiye's eligibility process, ÇÖZGER (Ministry of Health) and RAM (MoNE), produce categorical team-based determinations rather than continuous scores with nationally standardized cutoffs (Aydoğan Baykara et al., 2022; Resmî Gazete, 2019). This precludes comparison of children immediately above and below an eligibility threshold. The Alt 2 registry should therefore incorporate the Denver-II Turkish (Anlar & Yalaz, 1996) during RAM assessment as a standardized continuous baseline, the same instrument used in Türkiye's only RCT for this population (Karaaslan et al., 2013). Once two years of registry data are available and MoNE formally adopts a standardized score cutoff, this stronger causal comparison becomes feasible in Years 3–4.

Outcomes and Indicators

Primary outcomes	***Instrument***	***Timing***
School readiness at kindergarten	Denver-II Turkish (Anlar & Yalaz, 1996)	Month 36
Special ED placement rate at age 6	MoNE administrative records	Months 36–48
Secondary outcomes	**Instrument**	**Timing**
Child communication behavior	Denver-II Turkish (Anlar & Yalaz, 1996)	Months 0, 6, 12
Quality of parent-child interaction (responsiveness, affect, directiveness)	Turkish Version Maternal Behavior Rating Scale (TV-MBRS; Diken et al., 2009)	Months 0, 6, 12
Parent self-efficacy and wellbeing	Parent self-report (Renwick et al., 2003; WHO-UNICEF, 2023)	Months 6, 12
Child participation in daily activities	Parent-reported observation (Renwick et al., 2003)	Months 6, 12

Rationale. Denver-II Turkish and TV-MBRS enable direct comparability with Karaaslan et al. (2013), the only Turkish RCT for this population. Denver-II Turkish is validated for children aged 0–6 and feasible in semi-urban and rural settings (Eratay et al., 2015). TV-MBRS measures parent-child interaction quality, the core RT mechanism and has documented feasibility for community-based practitioners in Turkish settings (Çelen Yoldaş et al., 2020). Parent self-efficacy, wellbeing, and child participation in daily activities serve as Quality of Life outcomes, consistent with established evaluation criteria for this policy initiative (Renwick et al., 2003; WHO-UNICEF, 2023). The projected +0.15–0.22 SD benchmark provides a testable target, contingent on coaches delivering the program as designed and parents attending at least 75% of sessions.

Process outcomes Indicator	***Measurement***	***Timing***
Denver-II Turkish score	Administered at RAM assessment; recorded in registry	At enrollment
ÇÖZGER date	Date of medical eligibility determination	At enrollment
RAM assessment date	Date of educational placement determination	At enrollment

ÇÖZGER-to-RAM lag time	Weeks between eligibility and educational placement	Ongoing
Time from RAM to first service	Weeks from placement to first RT session	Ongoing
Dosage compliance	% of prescribed sessions attended per child	Monthly
Registry record completeness	% of required fields completed	Quarterly

Why ÇÖZGER-to-RAM lag time matters. Evidence suggests children in Türkiye are identified late, outside the 18–36 month window (Metin et al., 2025). Delays between assessments directly reduce available intervention time; tracking this lag from Month 1 identifies access barriers and triggers a navigation support response.

Process Evidence

Quantitative indicators. Program delivery fidelity: independent observers rate 30% of sessions per coach using the RT Intervention Session Guide (Karaaslan et al., 2013), verifying delivery of required RT session components. Fidelity benchmarks: 45–55% Year 1; 65–70% Year 3, reflecting current Turkish implementation capacity (Aksu, 2022) and trial conditions (Karaaslan et al., 2013). If Year 1 fidelity falls below 45%, scale-up is paused pending a coaching quality intervention. Additional indicators include session attendance per family, coach supervision hours per month, and registry completeness, dosage compliance, and ÇÖZGER-to-RAM lag rates (Alt 2).

Qualitative evidence. Structured interviews are conducted each round with parents, coaches, and center directors to capture variation by province type, disability category, and family background. Interviews address what is changing in daily parent-child interactions, what barriers prevent attendance, and whether families are being identified within the 18–36 month target window. A sample of 30% of each coach's RT sessions are video recorded and independently rated for parent-child interaction quality using the TV-MBRS (Diken et al., 2009), replicating Karaaslan et al.'s (2013) fidelity observation protocol.

Testing the mechanism. Karaaslan et al. (2013) found improvements in parent responsiveness and child engagement by Months 4–6; similar patterns are anticipated here. At Months 6 and 12, analysis examines whether improvements in parent responsiveness are associated with improvements in children's Denver-II Turkish developmental scores. If child gains occur without improvements in parent responsiveness, qualitative interviews identify alternative explanations, such as changes in school environment, family circumstances, or outside services.

Alternative explanations for observed gains. Families who enroll may have greater baseline motivation than non-enrollees, meaning gains could reflect participant characteristics rather than RT effectiveness. To address this, enrolled children are compared with registry-identified but not-yet-enrolled children; baseline differences are statistically controlled. Gains may result from regular parent-adult interaction rather than RT-specific mechanisms. Family attendance and coach fidelity are tracked independently. If attendance predicts outcomes but fidelity does not, contact time is the likely driver, triggering a coaching quality review.

Ethics and Equity

Consent and data protection. Written informed consent is obtained in Turkish and, where relevant, in Kurdish or Arabic. Families are informed that withdrawal does not affect their child's entitlements services. Records are stored in a secured portal with role-based access and unique child identifiers; cross-system linkage requires data sharing agreement between the Ministry of Health and MoNE.

Equity lens in sampling, monitoring and reporting. From Month 1, enrollment data are broken down by province, disability type, and key demographic characteristics. If rural or lower-income families are underrepresented relative to the estimated 153,000–291,000 eligible children (Demirci & Kartal, 2016; WHO-UNICEF, 2023), a Month 6 equity check triggers community health worker referral and tele-coaching expansion. Alt 3 is restricted to two urban regions in Years 1–3, as Türkiye does not yet have a trained itinerant inclusion support workforce at national scale (General Directorate of Services for Persons with Disabilities and the Elderly [EYHGM], 2020); rural children receive Alt 1 from Year 1. Rural expansion proceeds only if pilot regions demonstrate fidelity of at least 65% and positive school readiness trends, a specialist workforce pipeline is established, and registry data confirm sufficient rural DD/D identification rates. All outcome reports and findings are reported explicitly and shared with MoNE, the Ministry of Health, and civil society organizations representing DD/D families.

Closing the evidence gap. This evaluation will contribute rigorous quasi-experimental evidence on EI effectiveness for children with DD/D in a UMIC context, a setting currently underrepresented in the global evidence base, directly addressing the identified evidence gap and building the evidence base MoNE requires for future scaling decisions.

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Early Intervention Policy Reform for Children With Developmental Delays/Disabilities, Ages 0–5 in Türkiye

A Roadmap to a Coordinated Early Intervention System

- Türkiye has 2,505 Rehabilitation Centers (EYHGM, 2020)
- What Is Missing Is a Coordinated Early Intervention System (Diken et al., 2012)

Presenter: Gamze Venable

Date: April, 2026

Problem and Context

Most Children With DD/D Are Invisible to the System

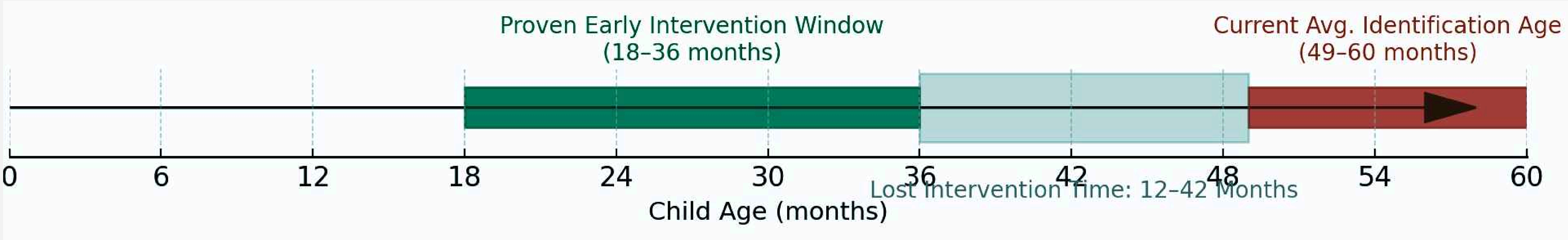
- Estimated 153,000–291,000 children ages 0–5 are eligible(Demirci & Kartal, 2016; WHO GBD 2019)

- Only 36,226 children are registered in the official disability registry (EYHGM, 2020).

- Estimated NOT registered: 116,774–255,000

How effectively does Türkiye's EI system identify and deliver quality services to all eligible children with DD/D both those in the registry and those who remain outside it?

Türkiye Is Systematically Missing the Critical Window
The Most Effective Intervention Years Pass Unaddressed



Children are identified at 49–60 months; the proven intervention window is 18–36 months, a system-wide delay of 12–42 months (Metin et al., 2025)

EI system is fragmented across three ministries (MoNE, MoH, MoFSS) with no standardized assessment, no outcome tracking, and shortage of trained specialists (Diken et al., 2012)

Access inequities are compounded by SES, geography, and disability type, unmeasured at national level(Diken et al., 2012)

Very few children are flagged early: Less than 1 in 150 children seen in pediatric clinics are referred for developmental checks (Metin et al., 2025)

MoNE = Ministry of National Education; MoH = Ministry of Health; MoFSS = Ministry of Family and Social Services

The Gap Is Structural: Infrastructure and Legal Mandate Already Exist, Coordination Does Not

Infrastructure

2,505 rehabilitation centers across all 81 provinces, grown 90% over 12 years, the delivery network for a scaled EI system already exists (EYHGM, 2020)

Legal mandate

The 2018 Special Education Regulation already requires ECI for ages 0–36 months and integrated preschool for ages 36+ months — no new legislation required (Aksu, 2022)

Coordination

Three ministries hold separate mandates with no shared data, no standardized service package, and no outcome tracking (Diken et al., 2012)

Key Evidence

Critical gap: No long-term Turkish outcome data exists

Turkish RCT Strongest evidence

Responsive Teaching → 42% DQ improvement vs. 7% in controls
Randomly assigned; fidelity monitored
(Karaaslan et al., 2013)

U.S. administrative cohort (N=214,370)

EI before age 3 → children scored slightly higher on Grade 3 reading
Benefit was stronger for highest-need children, ELA ratio 1.28 vs. 1.09 overall
(Stingone et al., 2026)

LMIC evidence

Early gains diminish when support stops enrollment alone is not enough, children need continued support through kindergarten entry (Jeong et al., 2021)

- U.S. National Cohort one contrary finding: EI recipients scored lower at kindergarten (−0.21 SD reading); reflects greater baseline need (Sullivan & Field, 2013)

Recommendation

Adopt a Hybrid Model: Three Components, Each Necessary, None Sufficient Alone

1

Parent-mediated Responsive Teaching (RT) the core intervention

Scale RT coaching nationally through existing rehabilitation centers for children 18–60 months, using hybrid in-person and tele-coaching (Karaaslan et al., 2013)

2

Minimum Service Package, registry, and family navigation as the accountability backbone

Define standards for services; build a national registry to track who is reached, with what intensity, and with what outcomes; deploy navigators to reduce access barriers (Sapiets et al., 2023)

3

Transition and inclusion pilot in two high-need regions

Use itinerant ECSE coaches to support preschool and kindergarten inclusion and transition. Provide monthly teacher coaching cycles and targeted small-group support (Aksu, 2022)

Each component is necessary; none is sufficient alone. RT delivers gains, MSP + registry makes them visible and equitable, and the inclusion pilot builds the missing evidence. Year 3 target: 8,000–10,000 children (3–7% of estimated eligible; Demirci & Kartal, 2016)

Criteria For Evaluating Policy Alternatives

Criterion	Alt 1 RT coaching	Alt 2 MiN. Service Package & Registry	Alt 3 Transition & Inclusion Pilot	Hybrid
Effectiveness	🟢	🟡	🟡	🟢
Equity	🟢	🟡	🔴	🟢
Scalability & Fidelity	🟢	🟡	🔴	🟢
Fiscal Cost	🟡	🟢	🔴	🟡
Political Feasibility	🟢	🟡	🟡	🟢
Quality of Life	🟢	🟡	🟡	🟢

Direct intervention without measurement produces unverifiable outcomes; measurement without intervention produces no child gains (WHO-UNICEF, 2023)

🟢 = Strong
🟡 = Partial
🔴 = Weak or absent

Sources: Karaaslan et al. (2013); Aksu (2022); Metin et al. (2025); EYHGM (2020); Renwick et al. (2003); WHO-UNICEF (2023)

The Hybrid Model Implementation

Policy Alternative	Months 0–6	Months 6–12	Years 2–3
Alt 1 RT Coaching	* Adapt materials; train coaches; select 4–6 provinces	Pilot in 4–6 provinces; fidelity target 45–55%	Expand to 81 provinces if fidelity ≥65%
Alt 2 Registry/MSP	* Define MSP standards; build registry; train staff	* Preparation to Month 9 Pilot Month 9	Expand nationally, strengthen registry system
Alt 3 Inclusion Pilot	Active in 2 urban regions	2 regions only; Month 6 equity check	Rural expansion only if all 4 conditions met

Active delivery *Preparation phase (will begin)

Conditional (only if conditions met)

Tradeoffs

Speed vs. accountability
Alt 1 and Alt 2 begin preparation together from Month 0 (Metin et al., 2025)

Urban pilot vs. rural inclusion
Rural children get Alt 1 from Year 1 (EYHGM, 2020; Aksu, 2022)

Manual tracking vs. digital
Basic system now; strengthen from Year 2 (Aksu, 2022)

Risks

Late identification persists
Track attendance and fidelity separately (Karaaslan et al., 2013)

Coach fidelity too low
30% session observation; pause if <45% (Aksu, 2022)

Gains reflect visits not RT method
Track attendance and fidelity separately (Karaaslan et al., 2013)

Ministerial data sharing
Agreement signed before Month 1

Evaluation Strategy

Three Strategies Generate Rigorous Evidence Without an RCT

- **Strategy 1 Staggered rollout**

Year 1 vs. Year 2 provinces; compare child scores and school readiness (Karaaslan et al., 2013)

- **Strategy 2 Before/after registry**

Track referral rates, ÇÖZGER–RAM lag times, and dosage compliance before and after registry activation (Metin et al., 2025)

- **Strategy 3 Matched comparison**

2 pilot regions vs. 2 matched non-pilot regions (Aksu, 2022)

Why not an RCT?

Three constraints:
Withholding services would violate legal and ethical obligations

Only the most visible DD/D cases are currently identified, representative randomization is not possible (Metin et al., 2025)

Registry is built during Year 1, no baseline exists for pre-assignment randomization

Four Questions Will Drive Scale-Up Decisions

Year 3 Is the First Defensible Evidence Point when first cohort reaches kindergarten

Are equity gaps narrowing?

Enrollment tracked by province, SES, and disability type from Month 1 Month 6 equity check → triggers response if rural/low-income families underrepresented

Does parent responsiveness drive child gains?

Instrument: Turkish Maternal Behavior Rating Scale at months 0, 6, and 12 If child gains occur at without parent improvement → investigate alternative explanations (Karaaslan et al., 2013; Diken et al., 2009)

Does the registry reduce system delays?

Track evaluation (ÇÖZGER–RAM) lag times and dosage compliance from Month 1 First answer: Year 1(Metin et al., 2025)

Does RT before 36 months improve school readiness?

Instrument: Denver-II Turkish First answer: Year 3 (Anlar & Yalaz, 1996; Karaaslan et al., 2013)

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